

Without Damage Function

$P = 150 - 17.21h$, $G(x) = gx (1 - \frac{x}{a})$ where a is the carrying capacity.

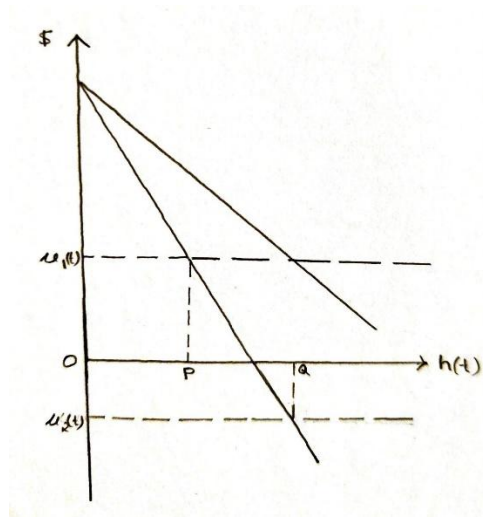
For max $J = \int_0^\infty p(h)he^{-rt}dt = \int_0^\infty (150 - 17.21h)he^{-rt}dt$
 $h(t)$

Subject to $\dot{x}(t) = G(x) - h = gx (1 - \frac{x}{a}) - h$, $x(0) = x_0$.

So, the current value Hamiltonian, $\mathcal{H} = (150 - 17.21h)h + \mu(gx (1 - \frac{x}{a}) - h)$.

(i): FOC: $\mathcal{H}_h = 0 \Rightarrow \mu = 150 - 34.42h$:

Since h here is the number of hunting permits sold, the First Order Condition says that the land owners should continue selling permits until the marginal revenue received from selling an additional permit $(150 - 34.42h)$ just balances the foregone value of an additional (in situ) unit of pig (μ). So, we need to find the number of hunting permits sold (h) that satisfies this equation overtime. Here, h and μ are negatively related and so as μ increases, h falls (if the value from preservation goes up, owners would want less hunting activities and more preservation).



Here, when μ is at $\mu_1(t)$ at the instance t , the optimal hunting is OP (after which the optimal hunting is zero), but as μ falls from $\mu_1(t)$ to $\mu_2(t)$, optimal hunting increases from OP to OQ (after which the optimal hunting is zero).

(ii): DC: $\dot{x}(t) = gx (1 - \frac{x}{a}) - h$, $x(0) = x_0$.

The Dynamic Constraint says that x behaves accordingly which is that the number of (in situ) wild pigs changes at the rate of their biological growth minus the hunting rate and initially (before hunting), the number of wild pigs is a constant, x_0 .

(iii) PB: $\dot{\mu}(t) = r\mu - \mathcal{H}_x = \mu (r - g (1 - \frac{2x}{a})) \Rightarrow \frac{\dot{\mu}}{\mu} + g (1 - \frac{2x}{a}) = r$:

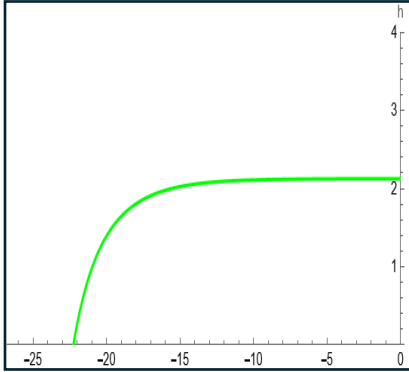
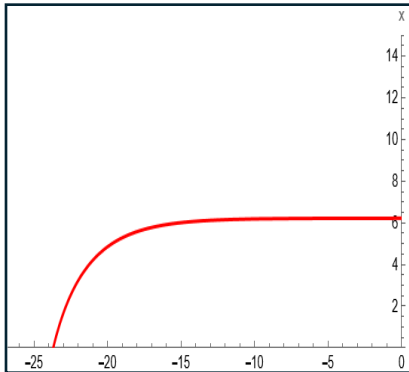
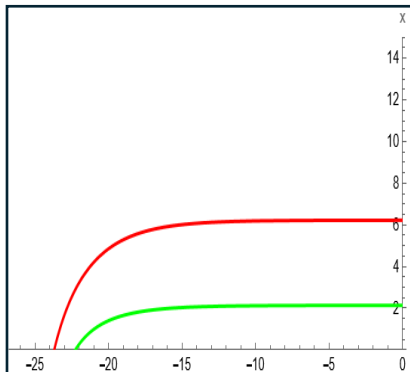
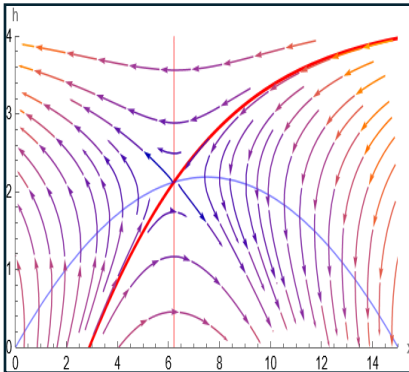
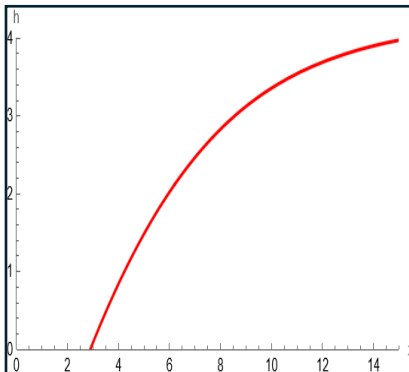
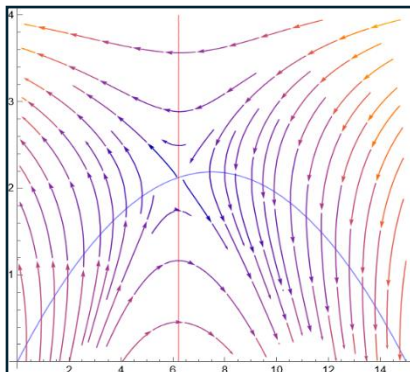
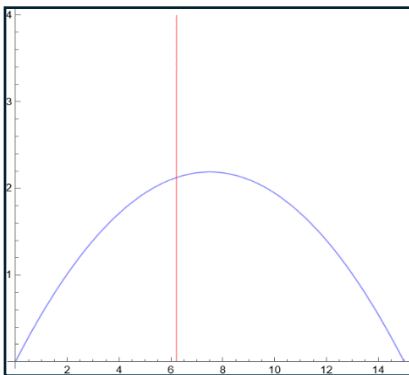
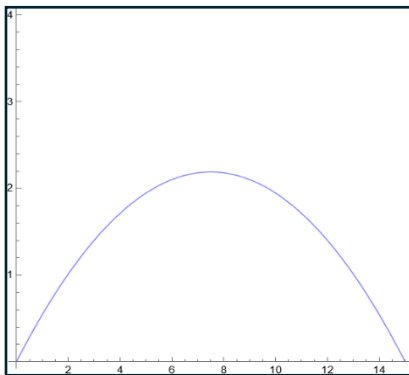
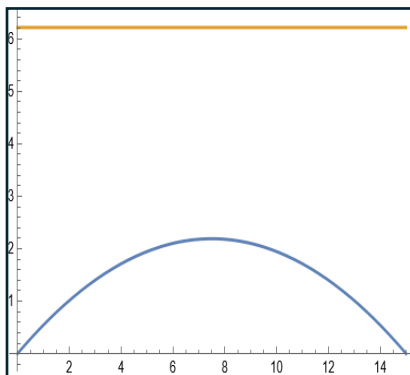
Here, in the left-hand side, $\frac{\dot{\mu}}{\mu}$ is the rate of capital gain/loss from holding one more pig in the wild given its current shadow value and $g(1 - \frac{2x}{a})$ is the rate of marginal net growth from holding on to an additional pig in the wild. Together, the left side of the equation represents the own rate of return (net benefit from conserving) from preserving an additional wild pig and the right side is the required rate of return for hunting. The Portfolio Balance says that we're indifferent between preserving and hunting the wild pig when the own rate of return balances the required rate of return (return from the natural asset balancing the potential return from other financial or physical assets if the owners sold the permits and invested in those external assets and gained returns from them). So, we need to find the number of pig (x) that satisfies this equation overtime.

Time differentiating (i) $\Rightarrow \dot{\mu} = -34.42\dot{h}$ (iv)

(i), (iii), (iv) $\Rightarrow (150 - 34.42h)(r - g(1 - \frac{2x}{a})) = -34.42\dot{h}$

$\Rightarrow \dot{h} = \frac{50}{1721}(34.42h - 150)(r - g(1 - \frac{2x}{a}))$ (v)

For $a = 15$, $r = 0.1$, $g = 0.5840$, from the Mathematica code, $(x^{ss}, h^{ss}) = (6.21575, 2.12579)$, the eigenvalues are $\lambda_1 = -0.369892$, $\lambda_2 = 0.469892$ (Saddle Point), and the diagrams are as below.



With Damage Function

With the damage function, $D(x) = cx$ where $c = 0.06$, $R = 100$,

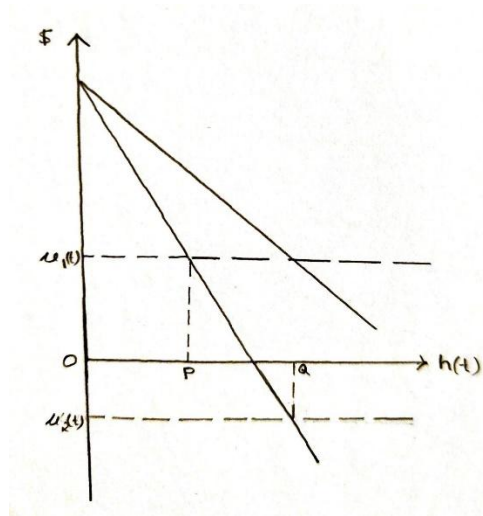
$$\max_{h(t)} J = \int_0^\infty (p(h)h + R(1 - D(x))e^{-rt} dt = \int_0^\infty ((150 - 17.21h)h + R(1 - cx))e^{-rt} dt$$

Subject to $\dot{x}(t) = G(x) - h = gx(1 - \frac{x}{a}) - h$, $x(0) = x_0$.

So, the current value Hamiltonian, $\mathcal{H} = (150 - 17.21h)h + R(1 - cx) + \mu(gx(1 - \frac{x}{a}) - h)$.

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(ii): DC: $\dot{x}(t) = gx(1 - \frac{x}{a}) - h$, $x(0) = x_0$:

The Dynamic Constraint says that x behaves accordingly which is that the number of (in situ) wild pigs changes at the rate of their biological growth minus the hunting rate and initially (before hunting), the number of wild pigs is a constant, x_0 .

(iii) PB: $\dot{\mu}(t) = r\mu - \mathcal{H}_x = \mu(r - g(1 - \frac{2x}{a})) + cR \Rightarrow \frac{\dot{\mu}}{\mu} - \frac{cR}{\mu} + g(1 - \frac{2x}{a}) = r$:

Here, in the left hand side, $\frac{\dot{\mu}}{\mu}$ is the rate of capital gain/loss from holding one more pig in the wild given its current shadow value, $g(1 - \frac{2x}{a})$ is the rate of marginal net growth from holding on to an additional pig in the wild and $-\frac{cR}{\mu}$ is the rate of marginal damage incurred by holding an additional

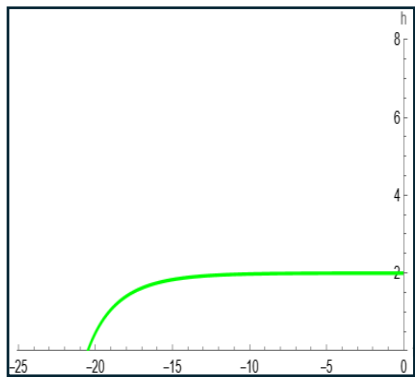
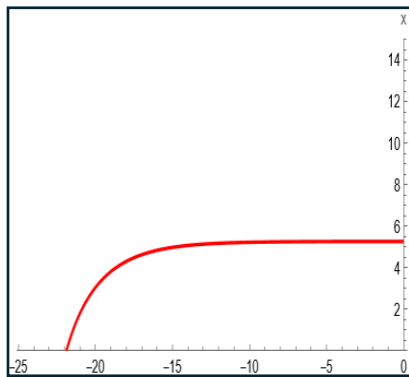
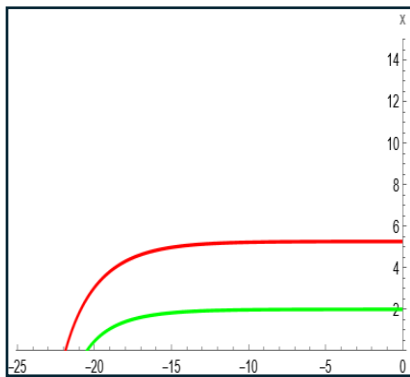
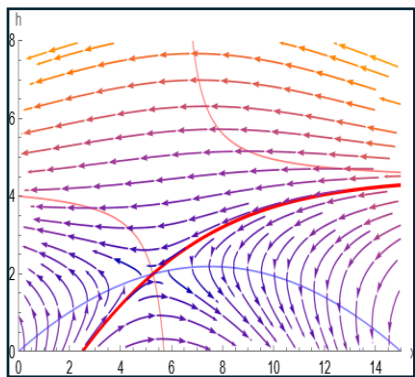
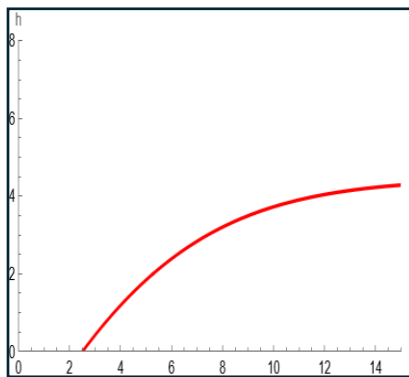
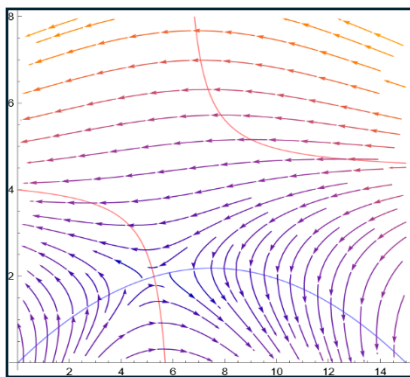
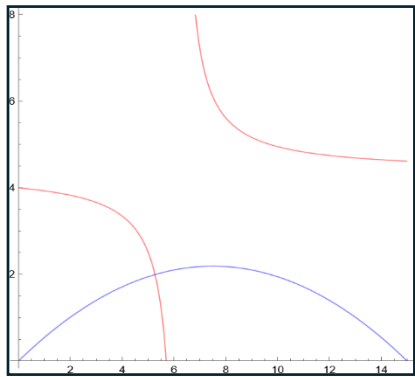
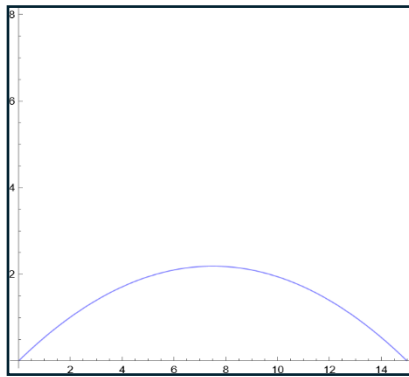
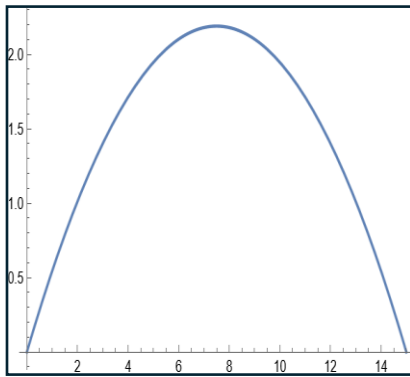
pig in the wild. Together, the left side of the equation represents the own rate of return (net benefit from conserving) from preserving an additional wild pig and the right side is the required rate of return for hunting. The Portfolio Balance says that we're indifferent between preserving and hunting the wild pig when the own rate of return balances the required rate of return (return from the natural asset balancing the potential return from other financial or physical assets if the owners sold the permits and invested in those external assets and gained returns from them). So, we need to find the number of pig (x) that satisfies this equation overtime.

Time differentiating (i) $\Rightarrow \dot{\mu} = -34.42\dot{h} \dots \dots \dots$ (iv)

$$(i), (iii), (iv) \Rightarrow (150 - 34.42h) (r - g (1 - \frac{2x}{a})) + cR = -34.42\dot{h}$$

$$\Rightarrow \dot{h} = \frac{50}{1721} ((34.42h - 150) (r - g (1 - \frac{2x}{a})) + cR) \dots \dots \dots (v)$$

Solving (ii), (v) simultaneously using Mathematica for $a = 15$, $r = 0.1$, $g = 0.5840$, $c = 0.06$, $R = 100$, we find $(x^{ss}, h^{ss}) = (5.26793, 1.99603)$, the eigenvalues are $\lambda_1 = -0.396364$, $\lambda_2 = 0.496364$ (Saddle Point), and the diagrams are as below.



In general, the stock effect would suggest that the additional pig population can increase agricultural damage, thereby resulting in higher costs for landowners. A decrease in the population may alleviate the damage and associated costs (increasing agricultural profit margin), but it might also lead to less revenue generated from selling hunting permits.

Here in this problem, when the damage to the agricultural revenue caused by the wild pig population is taken into account, preserving wild pigs become notably less attractive as it becomes urgent to take more necessary (trapping) steps to reduce the optimal amount of wild pigs preserved in the area rather than recreational ones (hunting), and so the optimal number of animals preserved falls and the optimal number of hunting also falls eventually. Another reason that might explain this is that when there are fewer animals preserved, optimal hunting activities have to fall to keep the dynamic balance. However, the FOC and the DC in this problem do not change due to the introduction of the damage function.

Another interesting point is that without considering the damage function, the h isocline was independent of h (vertical line), but as the damage function is considered, the h isocline becomes a piecewise (discontinuous) $1/x$ function as it has $h(t)$ in the denominator. However, the solution path still remains a saddle converging from the 1st and 3rd coordinates, and as we trace back from the boundary condition, it takes a shorter amount of time for x and h to reach their dynamically optimal levels when the damage is considered relative to when it's not. This tells us that when the landowners do not account for the damage, they are slow to respond (not fully recognizing the extent of agricultural damage initially) to economic incentives and environmental conditions, and they may underestimate the true costs initially, prioritizing short-term gains from issuing hunting permits without fully considering the long-term consequences of agricultural damage.